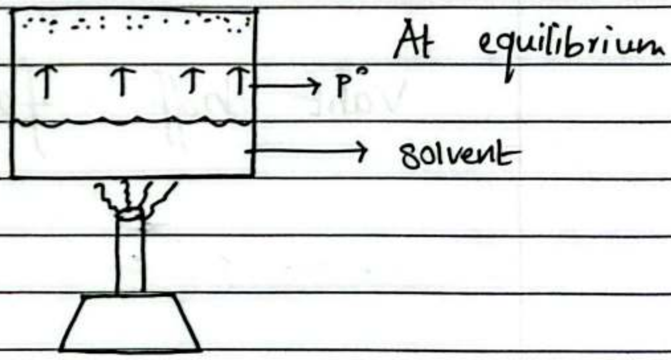
Lp~1

Solutions

— Colligative Properties
depends on number of moles of solute

1. Relative lowering of Vapour Pressure.
2. Elevation in Boiling point.
3. Depression in Freezing Point.
4. Osmotic Pressure.

Vapour Pressure ⇒



⇒ whenever a non-volatile solute is added to solvent, its vapour pressure will decrease

⇒ Raoults Law

For a Solution of volatile liquids, the partial vapour pressure is $\propto$

$$\text{Total Pressure} = P^\circ_A x_A + P^\circ_B x_B$$

$$P_i \propto x_i$$

$$\boxed{\frac{P^\circ_A - P_A}{P^\circ_A} = x_B.}$$

or

$$\boxed{\frac{P^\circ_A - P_A}{P^\circ_A} = \frac{n_B}{n_A + n_B}}.$$

or

$$\boxed{\frac{P^\circ_A - P_A}{P^\circ_A} = \frac{W_2 M_1}{W_1 M_2}}$$

P°_A = Pure non-volatile solvent Vapour Pressure.
P_A = Solution Vapour Pressure.
x_B = Mole fraction of solute.

Vaht hoff factor $\Rightarrow \frac{P^\circ_A - P_A}{P^\circ_A} = x_B \cdot i$

⇒ Elevation in Boiling Point.

Addition of solute increases boiling point

$$\boxed{\Delta T_B = \underbrace{T_B}_{\text{soln}} - \underbrace{T^\circ_B}_{\text{solvent}}}$$

$$\Delta T_B \propto \text{molality}$$

$$\boxed{\Delta T_B = K_b \cdot m}$$

Vant hoff factor ⇒ $\boxed{\Delta T_B = K_b \cdot m \cdot i}$

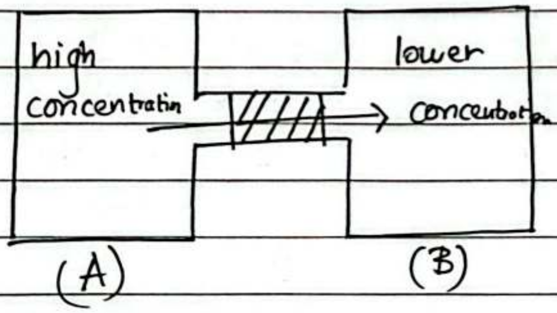
⇒ Depression in Freezing Point

$$\Delta T_f = T^o_f - T_f$$

$$\Delta T_f = k_f \cdot m$$

Vant hoff factor ⇒ $\Delta T_f = k_f \cdot m \cdot i$

⇒ Osmotic Pressure.



in kelvin

$$\pi = MRT \,/\, CRT$$

0.0821

molarity

$$\text{Molarity} = \frac{\text{moles of Solute}}{\text{volm in litres}}$$

Vant hoff factor ⇒ $\pi = MRT \cdot i$

⟹ VANT HOFF FACTOR

when an electrolyte solution is take it eithers

— Associates ⟹ $(i<1)$
— Dissociates ⟹ $(i>1)$

⟹ The amount by which it associates/dissociates is called the vant hoff factor (i)

Solute
- Strong Electrolyte — 100 % dissociates ⟶ ionic compound
- weak Electrolyte — ions & molecules > 50 % dissociates

$NaCl \longrightarrow Na^{+} + Cl^{-} \implies i = 2.$

$AlCl_3 \longrightarrow Al^{+} + 3Cl^{-} \implies i = 4.$

Dissociation ⟹ $\alpha = \frac{i-1}{n-1}$

Association ⟹ $\alpha = \frac{i-1}{\frac{1}{n}-1}.$

⟹ $i = \frac{\text{Normal mol. mass}}{\text{calculated mol. mass}}$ or $\frac{\text{observed colligative property}}{\text{normal colligative property}}$

or

$i = \frac{\text{Total moles after asso/disso}}{\text{Total moles befor asso/disso}}$

IMPORTANT COMPOUNDS for Solutions Chapter.

⇒ Important Compound masses.

i] Water (H_2O) — 18
ii] Glucose ($C_6H_{12}O_6$) — 180
iii] Urea (CH_4N_2O) — 60
iv] Sucrose ($C_{12}H_{22}O_{11}$) — 342
v] Acetic Acid (CH_3COOH) — 60
vi] Ethanol (C_2H_5OH) — 46
vii] Benzene (C_6H_6) — 78
viii] Napthalene ($C_{10}H_8$) — 128
ix] Camphor ($C_{10}H_{16}O$) — 152
x] Cyclohexane (C_6H_{12}) — 84
xi] NaCl (NaCl) — 58.5
xii] KCl — 74.5
xiii] $CaCl_2$ — 111
xiv] $MgSO_4$ — 120
xv] Na_2SO_4 — 142
xvi] $AlCl_3$ — 133.5

⇒ i values for common electrolytes

Glucose — 1
NaCl — 2
KCl — 2
HCl — 2
$MgSO_4$ — 2
$CaCl_2$ — 3
Na_2SO_4 — 3
K_2SO_4 — 3
$AlCl_3$ — 4

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